A REPORT

ON

ADVANCEMENTS IN AUXILIARY AND SULPHURIC ACID PLANT

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Grasim Industries Nagda

A Practice School-I Station of

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

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Chapter -1

Improving Salt Whiteness

Introduction

The present report constitutes a mid-term review of the Practice School project, aimed at increasing the whiteness of salts, mainly sodium sulphate, recovered from Grasim Industries' auxiliary and sulphuric acid plants. The project safeguards operational effectiveness and product quality, in line with Grasim's sustainability and Zero Liquid Discharge (ZLD) philosophy.

Current Practices & Challenges

Existing Salt Recovery Process

- Effluent Treatment: Released water and salts are recovered from effluent streams by multi-effect evaporators (MEE), reverse osmosis (RO), and crystallizers.
- Pre-Treatment: Coagulation and flocculation with alum, PAC, ferric chloride, followed by sand/multimedia filtration to remove suspended solids and some dissolved impurities.
- Brine Concentration & Crystallization: The brine undergoes concentration through forced circulation and draft tube baffle (DTB) crystallizers, after which salts are obtained by centrifugation and drying method.

Quality Control and Limitations

- Discoloration Problems: Residual organics along with metal ions remain unnoticed in the conventional processes and cause the yellow or grey discoloration of the salt.
- Outdated Processes: advanced chemical bleaching or ac

Proposed Advancements for Enhancing Whiteness

Enhanced Purification & Filtration

- Optimized Coagulation and Flocculation: A real-time pH-and-turbidity instrument must be used to analyze and optimize dosages, thereby improving the removal of color bodies and organics.
- Mandatory Microfiltration and Ultrafiltration: MF and UF processes must be considered standard treatment steps (not optional) to ensure the continuous removal of fine particulates and dissolved organics, which impart color impurities.

Advanced Brine Treatment

- Activated Carbon Filtration Post-RO:Activated carbon beds should be installed downstream of RO systems. These will help in adsorption of trace colorants, and regular regeneration which ensures effectiveness.
- Chemical Bleaching: Use stable bleaching powder/pilot ozone/hydrogen peroxide treatments during salt washing for the oxidation of color-causing compounds.

Process Parameter Optimization

- Regression Analysis for Process Control: Use regression methods to set ripening and spinning line temperature parameters that maximize yield and thereby improve salt whiteness.
- Real-Time Dashboards: Deploy a dashboard to monitor all salt quality KPIs (color, iron content, turbidity) and act quickly upon deviations.

Auxiliary Plant Improvements

- Multi-Stage Filtration & Ion Exchange:An ion-exchange brine purification system must be installed in auxiliary plants to remove the rest of the impurities.
- Improved Crystallization & Washing:The crystallization and washing steps must be improved to produce crystals of uniform shape and purer quality.
- Enhanced Drying & Handling:Better drying and gentle handling will keep the salt white by not allowing recontamination of the salt.

Impact and Safety

Expected Impact

- Consistent Salt Whiteness: Enhancing the purification and whitening stages will yield salts with a better degree of whiteness and more uniform purity.
- Reduced Operational Variability:Real-time analysis and process control will work toward reducing any fluctuations in salt quality.
- Improved Process Efficiency: With better process optimization and integrated technology, great opportunities for upgrading the process efficiency at a big scale are unveiled.

Safety & Handling

- Strict Chemical Handling: Necessary precautions need to be taken concerning handling acids, caustics, and oxidants to avoid contamination and, secondly, from the standpoint of safety and quality of the final product.

Conclusion

With the combination of high-filtration processes, chemical whitening, analytical processes, and parameter optimization, Grasim's auxiliary and sulphuric acid plants shall establish the new standard of quality for salt; therefore, this will help toward operational excellence and sustainability, further putting Grasim at the forefront of eco-efficient industrial salt production.

Chapter - 2:

Comparison Between Acid Based and Steam Based Crystallizer

Introduction

As part of my internship at Grasim Industries Nagda, a flagship unit of the Aditya Birla Group and one of the major producers of viscose staple fibre (VSF) — I am assigned a project entitled "Advancements in Auxiliary and Sulphuric Acid Plant." Under this broader project, my sub-project deals with the comparative study between acid-based and steam-based crystallizers, which are an essential part of the sulphuric acid plant.

The objective of this research is to examine both technologies in such crystallization processes that help to enhance plant productivity, sustainability, and product quality. Crystallizers play a vital role in by-product recovery and purity level required for further processing. Design and operations of crystallizers affect several parameters such as energy consumption, operational safety, efficiency, and environment.

The project involves an in-depth study of the working principles, advantages, disadvantages, and industrial application of both type crystallizers. Based on my analysis of how they work relative to conditions and demands of Grasim, through identification of areas for improvement, recommendations on the most suitable crystallization technology for future process development can be made

Importance of crystallizers in Sulphuric Acid Plants

Byproduct recovery is very important part of the viscose manufacturing process, especially through the recovery of compounds of value such as sodium sulphate (Na_2SO_4). This contributes to both the economic and added value but also significantly towards the aim of sustainability. Recycling and recovery of by-products assist in reducing wastage by decreasing the amount of waste material sent to landfill, quite in accordance with the tenets of a circular economy. Further, the operation optimizes overall efficiency by reducing thermal energy wastage and minimizing use of fresh raw materials, thereby realizing cost savings and maximum utilization of resources. In addition, successful byproduct recovery aids in environmental compliance with the potential to embrace practices such as zero-liquid-discharge (ZLD) and the reduction of toxic sulphurous emissions, thereby ensuring compliance with regulatory needs and promoting eco-friendly operations.

Acid-Based Crystallizer

This sophisticated crystallization system leverages the heat produced exothermically during Double Contact Double Absorption (DCDA) process and thus is a very energy-efficient process for the recovery of valuable by-products. It converts the thermal energy produced from the absorption of sulphur trioxide (SO_3), which can range from 80–120°C, and processes under vacuum to maximize energy consumption. In doing so, sodium sulphate is recovered as Glauber's salt ($Na_2SO_4 \cdot 10H_2O$) using a multi-effect evaporation (MEE) plant and then converted to its anhydrous form for subsequent applications. For withstanding the corrosive environment, the system utilizes high-quality, corrosion-resistant material like Hastelloy and titanium, which makes the system long-lasting and dependable.

One of the most significant benefits of this design is that it can decrease the requirements for external energy by up to 30–40%, and that contributes significantly to decreasing the cost of operations. Furthermore, the process is also marked by having a very high recovery of over 98% for sodium sulphate and recycles between 95% of process water, which is a testament to how energy-efficient it is. From a sustainability point of view, the system relates to a captive power plant, which serves to minimize the use of coal by about 15–20%, also decreasing the carbon footprint of the operation. The system is, however, confronted with some operational issues, such as the precipitation of sodium sulphate within the MEE units, requiring the systematic application of EDTA to clean and maintain. Additionally, employing high-quality materials and technological infrastructure yields a high initial capital outlay, which, even though it has long-term advantages, can present a hindrance to large-scale application.

Steam-Based Crystallizer

This system efficiently utilizes waste heat produced during the burning of sulphur in the Double Contact Double Absorption (DCDA) system, in which the sulphur is burned at very high temperatures of around 1000°C. Such intense heat energy evolved during the reaction is utilized to produce steam using waste heat boilers, which usually produces steam at the temperatures of 173°C to 187°C. The steam is also a useful source of energy for other downstream operations such as crystallization. The equipment is built mainly with carbon steel, which is cost-efficient and safe and suitable for steam application. One of the advantages of this method is its capability to greatly enhance energy efficiency by reclaiming wasted heat otherwise, hence lessening the consumption of coal in the plant by about 15–20%.Other than energy saving, the system has optimal efficiency in byproduct recovery, especially sodium sulphate, and permits recycling of over 95% of process water, saving resources in the process. Environmentally, the process minimizes sulphurous emissions and compliments overall sustainability goals, such as the expansion of the production plant to a capacity of 219,000 tonnes-per-annum (TPA). Economically, the process is associated with considerable cost savings through the use of internal heat streams and avoiding the use of external fuel resources and minimizing operating costs in the long term.

Comparison between the two

The acid-based and steam-based crystallizers are applied in sodium sulphate recovery and water recirculation, yet they are vastly different in their operational characteristics, material, and energy source.

The acid-based crystallizer taps the exothermic heat produced in SO_3 absorption in the DCDA process as its energy source. This renders it highly energy-efficient, cutting down on external energy supply by about 30–40%. The operating temperature of this system is between 80 to 120°C, which is adequate for the materials used—namely, corrosion-resistant alloys such as Hastelloy and Titanium, which provide robustness under the acidic condition. It has a sodium sulphate (Na_2SO_4) recovery rate of around 98% and recycling 95% of the process water, with high contributions towards resource efficiency. It does have disadvantages like high capital investment and problems like scaling of the tube in the Multiple Effect Evaporator (MEE), which can influence long-term performance and maintenance. It is typically integrated with the captive power plant facility to achieve optimization of energy utilization in the facility.

Conversely, the crystallizer that uses steam depends on recovered waste heat from combustion of sulphur, especially from the hot exhaust gases produced through sulfuric acid production. It runs at a higher temperature range between 173 and 187°C. Equipment in this system is usually carbon steel, which, though less resistant to corrosion than equipment used in acid-based systems, is cheaper and able to withstand the higher temperature and lower corrosiveness. Though the actual recovery of sodium sulphate is not always measured, it is thought to be high, and recycling of water is over 95%, indicating its efficiency. With regards to energy efficiency, the steam-crystallizer ensures the consumption of coal by around 15–20%, contributing to universal sustainability. However, this system also has its operation challenges, especially with respect to maintaining the best temperature control and balancing the maintenance requirements of heat exchangers. It is generally integrated into the sulfuric acid process directly, utilizing waste energy streams for increased efficiency.

In summary, the two crystallizer systems each possess advantages and disadvantages in aspects of efficiency, integration, materials, and operational issues. The selection of either of the two mainly relies on the industrial configuration, energy integration objectives, and long-term economic interests.

Chapter 3

Cooling Tower Efficiency Improvement

Introduction

As part of the internship at Grasim Industries Limited, Nagda, a flagship unit of the Aditya Birla Group, I have been assigned to a project focusing on the "Advancements in Auxiliary and Sulphuric Acid Plant." Under this broader project, my sub-project deals with “The Cooling Tower’s Efficiency Improvement”, which is an essential part of the sulphuric acid plant.

Role of cooling towers

Cooling towers are crucial in sulfuric acid production units, particularly in the contact process, where they maintain optimal temperatures for efficient and safe operations by dissipating the excess heat produced during the process.

Unlike natural heat sinks (lakes, rivers, etc.), cooling towers offer a much more efficient, consistent, and controllable method of cooling and prevent excessive heat release into the environment regardless of any variations in temperature, humidity, etc.

Uncontrolled temperatures can cause thermal stress in reactors, towers, and piping, leading to material fatigue and failure. By maintaining consistent temperatures, cooling towers extend the lifespan of such equipment.

In the DCDA (Double Contact Double Absorption) process for sulphuric acid production, cooling towers can be found in the following steps:

- Cooling towers help maintain optimal operating conditions for catalysts. The oxidation of sulfur dioxide to sulfur trioxide, using a vanadium pentoxide catalyst, is most efficient between 400-450°C. Without proper cooling, excessive heat can reduce the catalyst's efficiency and lifespan.

- Controlling the temperature in the absorption process. In the absorption tower where sulfur trioxide and sulfuric acid produce Oleum, which is then diluted with water to give back sulfuric acid. This process releases significant heat that must be managed to prevent efficiency losses and equipment damage.

- Additionally, cooled water from the cooling tower is used for condensation and acid circulation in various production stages, ensuring stable acid concentration. It also cools the circulating acid, preventing overheating, corrosion, or unwanted side reactions.

- Lastly, they contribute to environmental and safety compliance. Cooling prevents excessive heat release into the environment, reducing the risk of thermal pollution. Proper temperature control also minimizes the release of acidic vapors, enhancing worker safety.

Working of a cooling tower

The core principle behind cooling tower operation is evaporative cooling. When hot water comes into contact with air, a small portion of the water evaporates, and this evaporation removes a significant amount of heat due to the high latent heat of vaporization of water. The remaining water is cooled as a result of this heat removal process.

Key Components and Their Functions;

- The **Fill Media** is the component designed to promote maximum heat transfer. It does this by ensuring maximum contact between air and water, creating a turbulent air-water flow regime, and slowing down water flow to increase contact time with air. The two primary types are splash fill and film fill, each designed to optimize different aspects of heat transfer. Splash-type fill can be made of wood or plastic. This type uses staggered bars or slats to break up the water flow, creating splash and droplets that enhance air-water contact. Film-type fill uses thin sheets of material and is generally made of polyvinyl chloride (PVC) with a flat, corrugated, or textured surface. The water spreads across these sheets in a thin film, maximizing the surface area for evaporation. The advantage of PVC as a material is that it is naturally fire resistant, whereas other plastics must be specially treated.

- The **Water Distribution System** ensures uniform water flow across the fill media. Hot water from the process enters at the top of the tower and is distributed through spray nozzles.

- **Drift Eliminators** capture large water droplets from the discharge air stream by forcing air and droplets to change direction suddenly, causing water to condense and fall back into the tower, thus helping meet the environmental regulations for water conservation.

- **Fans and Air Movement Systems** move large volumes of air efficiently through the fill media by creating the necessary air-water contact for heat transfer. They can be positioned either to push air into the tower (forced draft) or pull air through it (induced draft).

- The **Collection Basin** at the bottom of the tower collects the cooled water and serves as a reservoir before it returns to the process.

Improvement Strategies

Ensure the thermal performance is efficient and reliable by addressing the four primary concerns: *corrosion, scaling, fouling, and microbial growth*. To account for these concerns, regularly inspect and clean fill media, nozzles, and basins with chemical additives such as corrosion inhibitors, scale inhibitors, dispersants, and biocides to keep the cooling system surfaces protected and clean.

Using high-efficiency Fiber Reinforced Plastic (FRP) blades and variable frequency drives (VFDs) to match fan speed to actual cooling demand.

The thermal efficiency and longevity of the cooling tower and equipment depend on the proper management of recirculated water. Water leaves a cooling tower system in one of four ways;

1. **Evaporation** - The primary method used to transfer unwanted heat from the water into the environment.
2. **Drift** - Water droplets that escape through the top of the cooling tower or from the side in crosswinds.
3. **Blowdown** - Water that's discharged to remove impurities, sediment, and high mineral content from the supply. Depending on the amount of water required for proper cooling capacity, you can treat the blowdown for efficient reuse in the tower. Common treatments include reverse osmosis for filtration or ion exchange for softening. This way, unwanted contaminants are concentrated and removed from the water.
4. **Basin Overflows** - Properly running cooling towers will not experience leaks or overflows in the system. However, this type of water loss is equally important when calculating the amount of makeup water required.

The quality of the cooling tower feed water may indicate an abundance of silica or a need for pH stabilization. In this step of the cooling system process, water treatment typically involves technology that can help adjust the pH or remove hardness in the water. Chemicals are introduced at later points of the process, though they may not be enough for the bleed to drain. Proper feed water treatment can minimize the water bleed rate to the drain and optimize the tower evaporation cycles. The circulatory water within the tower may require some form of side-stream filtration to help retain a healthy balance. Chemical treatments typically occur at this point in the tower cooling process to minimize scale formation and prevent the corrosion of critical equipment. By running some of the circulated water through a side stream filter and introducing proper chemical treatment, the suspended particles and solids are dispersed instead of being left to settle on the surface.

All losses in the water supply must be replaced by makeup water. Tracking when and how much water your system loses is essential for water conservation.

A key parameter used to evaluate cooling tower operation is the ‘cycle of concentration’, determined by calculating the ratio of the concentration of dissolved solids in the blowdown water to the make-up water. A high cycle of concentration will minimize blowdown water quantity and reduce make-up water demand.

Install a make-up water or side-stream softening system when hardness (calcium and magnesium) is a limiting factor on cycles of concentration. Water softening allows you to operate at higher cycles of concentration.

Install covers on open decks, basins of the tower. Reducing the amount of sunlight on tower surfaces can significantly reduce biological growth, such as algae.

Cooling Tower Comparison

The Sulphuric acid production plant at Grasim Industries Ltd, Nagda employs the widely used Paharpur Cooling Towers. Paharpur Cooling Towers Limited is recognized as the world's largest manufacturer of process cooling equipment, with a legacy dating back to its founding in 1948 in Kolkata, India. The company has a global presence, exporting to over 50 countries and serving a wide array of industries.

Considering that the hardness of water is a significant issue at plants, cooling towers that specialise in water softening methods achieve the following;

- Material Selection - FRP (Fiberglass Reinforced Plastic) Cooling Towers. FRP is highly recommended for hard water environments because it offers excellent corrosion resistance, especially compared to metals that are more susceptible to scaling and corrosion from minerals in hard water. FRP towers are widely used in industries with challenging water quality, including chemical plants.

- Water treatment - Regular monitoring and adjustment of pH, alkalinity, and hardness to minimize scaling and corrosion, use of scale and corrosion inhibitors specifically formulated for hard water, such as ‘Tower Treat H’, which helps prevent and remove scale, lime, and mineral deposits. Advanced side-stream filtration and plasma water technology can further reduce limescale formation and corrosion, improving efficiency and reducing maintenance needs.

- Automated Controllers - Installing controllers that monitor conductivity and automate chemical dosing can help maintain optimal water chemistry, preventing scale buildup even when hard water is used.

If feasible, pre-treating the makeup water with lime softening, ion exchange, or reverse

osmosis can reduce hardness and allow the cooling tower to operate at higher cycles of concentration, improving efficiency and reducing water waste.

Here are a few competitors to Paharpur cooling towers in India;

1. Artech Cooling Towers Pvt. Ltd.
   Based in Gujarat, Artech manufactures a wide range of cooling towers, including Induced Draft FRP (Fiberglass Reinforced Plastic). Their products are tailored for industrial applications and can be customized to meet the specific needs of chemical and process plants, making them suitable for sulphuric acid production environments.
2. Siddhant Equipments Pvt. Ltd. (Aqua Cool Cooling Towers)
   Siddhant Equipments offers FRP Cooling Towers, Evaporative Cooling Towers, and both Natural and Mechanical Draught designs. Their towers are engineered for durability, energy efficiency, and can be customized for heavy industrial use.
3. Towertech India
   Towertech specializes in industrial water cooling towers used across sectors such as petrochemicals, oil refining, and manufacturing.
4. Classik Cooling Towers
   Classik manufactures a broad range of cooling towers, including FRP, wooden, adiabatic, closed-circuit, and natural draft towers. As a CTI-certified company, they offer solutions designed for aggressive industrial environments with expertise in custom engineering.

Chapter - 4

Advancements in H2SO4 Manufacturing process

Introduction

As part of internship at Grasim Industries Limited, Nagda, a flagship unit of the Aditya Birla Group, I am assigned to a project entitled "Advancements in Auxiliary and Sulphuric Acid Plant." Under this broader project, my sub-project deals with the Advancements in H2SO4 manufacturing process, which are an essential part of the sulphuric acid plant.

What is the H2SO4 manufacturing process?

It involves the contact process, a multi-step process where sulfur dioxide (SO2) is oxidized to sulfur trioxide (SO3), which is then reacted with concentrated sulfuric acid to produce oleum, which is subsequently diluted to form sulfuric acid.

Procedure:

1. Preparation of Sulfur Dioxide (SO2):
2. Conversion of SO2 to SO3 (Contact Process)
3. Absorption of SO3 and Production of Oleum
4. Dilution and Production of Concentrated Sulfuric Acid:

Raw materials required for manufacture of H2S04 process:-

1. Sulphur or Sulphur-containing Compounds

Elemental sulphur (S) – Most common, obtained from natural deposits or as a by-product from petroleum refining and natural gas processing.
Hydrogen sulphide (H_2S) – From petroleum and natural gas industries.
Metal sulphide ores – Like **pyrite (FeS_2)**; used less frequently now.

2. Air (Oxygen Source)

Required to convert **sulphur to sulphur dioxide (SO_2)** and **SO_2 to sulphur trioxide (SO_3)**.
Only the **oxygen (O_2)** from air is used in the chemical reactions.**3.Water (H_2O)**

Used in the final step to **absorb SO_3** and form sulphuric acid:
SO3+H2O→H2SO4$SO_3 + H_2O$ H_2SO_4SO3+H2O→H2SO4

Klaus Kiel method for sulphur extraction

A popular industrial technique for extracting elemental sulfur from hydrogen sulfide (H2S) gas, a byproduct of refining operations, particularly in oil refineries and natural gas processing facilities, is the Claus process.By lowering sulfur emissions and creating valuable elemental sulfur, this process is crucial for protecting the environment.

1. Partial Combustion (Thermal Stage):

- In a furnace, a stream of gas rich in H2S is combined with oxygen or air and burned.

- A portion of the H2S is converted to water vapor and sulfur dioxide (SO2) during this combustion.The reaction typically follows these equations:H2S + 1 ½ O2 → SO2 + H2O 2 H2S + SO2 → 2 H2O + 3 S (elemental sulfur)

- The hot gas stream is subsequently cooled in a waste heat boiler after producing a sizable amount of elemental sulfur at this stage.
- The waste heat boiler recovers heat from the hot gas, often producing steam.
- Condensed sulfur is extracted from the gas stream in the boiler.
- This reaction occurs at ~1000–1400°C in a combustion furnace.
  Produces a gas mixture containing sulfur vapors**,** SO_2, H_2S, and water vapor.
  - H2S + 1 ½ O2 → SO2 + H2O
  - 2 H2S + SO2 → 2 H2O + 3 S (elemental sulfur)

2. Catalytic Stage:

- The gas mixture is cooled to condense some of the sulfur.
- To further convert H_2S and SO_2 to sulfur, the residual gases travel through catalyst beds, which are typically made of titanium dioxide or alumina, at temperatures between 200 and 350°C:
  2H2S+SO2→catalyst 3S+2H2O $2H_2S + SO_2$->{catalyst} 3S + $2H_2O$
  2H2S+SO2catalyst3S+2H2O
- typically uses two or three catalytic converters, with each stage followed by cooling and sulfur condensation.
- Typically, alumina (Al2O3) is used as a catalyst.
- In order to produce more elemental sulfur, the catalyst facilitates the reaction

between H2S and SO2.

3. Sulfur Condensation:

- Elemental sulfur is condensed from the gas phase and collected as a liquid.

Overall Stoichiometry:

$3H_2S + 3/2O_2 \rightarrow 3S + 3H_2O$

Advantages of Claus Process:

- Extremely effective sulfur recovery (>99% with tail gas treatment; 95–97% in 2-stage systems).
- Reduces SO_2 emissions while adhering to environmental regulations.

Advancements in H2SO4 manufacturing process

The main focus of improvements in H2SO4 (sulfuric acid) production is on making the contact process more efficient. This process involves turning sulfur dioxide into sulfur trioxide and then mixing it with water. Double contact double absorption (DCDA) and wet sulfuric acid (WSA) processes are the most common, and DCDA has a higher conversion efficiency. Better catalysts, better plant equipment, and better modeling methods have all helped to boost production capacity and purity.

1. Enhanced Process Optimization:

DCDA Process:

This process has two steps: oxidation and absorption of sulfur trioxide. It has higher conversion efficiencies (over 99%) than single absorption processes.

Modeling and Simulation:

Aspen Plus and other tools are used to simulate and improve the contact process, which lets you control things like the flow rate of water very precisely to get the most H2SO4 production and purity.

Catalyst Optimization:

Vanadium pentoxide (V2O5) is an important catalyst in the contact process, and more research is being done on catalyst materials and how well they work.

2. Increased Production Capacity and Purity: DCDA Technology: DCDA

plants can make a lot of stuff, like sulfuric acid monohydrate (Mh), which can be up to 6,000 tons per day.

Optimized Plant Equipment:

Production per unit of chamber space has increased as a result of advancements in plant equipment, such as Mills-Packard chambers and intermediate towers in chamber plants.

Purity Enhancement:

High purity levels, such as 98.9% in certain situations, can be achieved in H2SO4 production through process optimization.

3. Alternative Sourcing and Process Variations:

Wet Sulfuric Acid Process (WSA):

The Wet Sulfuric Acid Process (WSA) provides an alternate way to produce H2SO4, especially for specific industrial uses.

Elemental Sulfur Utilization:

To further maximize the production of H2SO4, elemental sulfur may be investigated as a feedstock.

4.Automation and Process Control

Use of DCS (Distributed Control Systems) for:
Real-time gas concentration monitoring.

Optimized temperature control in converter beds.

Leak detection and emergency shutdown systems.

5.Improved Catalysts

Traditional Catalyst: Vanadium pentoxide (V_2O_5) supported on silica is a conventional catalyst.

Advancements:

To boost activity and lower the SO_2-to-SO_3 conversion temperature, promoters such as potassium sulfate are added.Higher surface area catalysts, also known as nano-catalysts, for increased efficiency and reduced operating temperatures.

6.Research in Alternative Methods

Growing interest in plasma-based and electrochemical processes for more environmentally friendly H_2SO_4 production

Major Equipment and Their Common Suppliers

Equipment/System	Typical Suppliers (Global & Indian)
Sulfur burners / Combustion system	MECS (DuPont), Monsanto, Chemetics, UHDE, Jacobs, Hindustan Dorr-Oliver
Catalyst beds (Vanadium Pentoxide)	MECS (DuPont), Topsoe, Chemetics, Heraeus
Heat exchangers / Waste heat boilers	L&T, BHEL, GEA, Thermax, Alfa Laval, HRS Process Systems
Acid coolers and pumps	KSB Pumps, Sulzer, Flowserve, Grundfos, Pfaudler, CP Pumpen AG
Drying / Absorption Towers	HDO (Hindustan Dorr-Oliver), Jacobs, Thermax
Mist eliminators / Demisters	Kimre Inc, Munters, Envitech, Gore-Tex, Sulzer Chemtech
Tail gas scrubbers	AirScience, Amiya Engineering, CECO, Triveni Engineering
Instrumentation & Control (DCS/PLC)	Siemens, ABB, Honeywell, Emerson, Schneider Electric
Gas cleaning system	Thermax, HDO, Lurgi, Chemetics, Envirotech
Storage tanks & acid resistant lining	Pfaudler, Kirloskar, Hindustan Composites, TFL

Chapter - 5:

Advancements in MSFE heater cleaning process

Introduction

As part of internship at Grasim Industries Limited, Nagda, a flagship unit of the Aditya Birla Group, I am assigned to a project entitled "Advancements in Auxiliary and Sulphuric Acid Plant." Under this broader project, my sub-project deals with the Advancements in MSFE heater cleaning process, which are an essential part of the sulphuric acid plant.

Multi-Stage Fusion Evaporator

The heater unit is used in a Multiple-Effect Evaporator (MSFE) system. It is essentially a shell-and-tube heat exchanger used to heat the feed liquid before it enters each evaporator effect.

Function:

An MSFE system uses steam to evaporate water or solvent from a feed liquid in multiple stages ("effects") to increase energy efficiency. The MSFE Heater plays a crucial role in this process.

1. Heats Feed Liquid:
   Steam is passed through the shell side of the heater.
   The feed liquid passes through the tube side.
   Heat is transferred from the steam to the liquid.

2. Prepares for Evaporation:
   The heated liquid is then sent to the evaporator body, where it boils under vacuum, separating water as vapor.
   Heating is done just below the boiling point at reduced pressure, minimizing energy use.

3. Repeats Across Effects:
   In later effects, vapor from the previous stage is reused as the heating medium, and a new heater is used in each stage.

At Grasim Industries, especially in VSF and chemical recovery, MSFE heaters:

- Concentrate chemical effluents or weak acid streams
- Improve steam economy (reduce fresh steam use)
- Handle high-TDS, corrosive feed liquids, requiring durable materials (titanium)
- Are critical in minimizing wastewater volume and energy consumption

Control Measures adopted MSFE RO feed increased for more water recycling resulted in water saving of 600 m3/day

Why is Cleaning Important?

Because the feed liquid can cause scaling and fouling inside the tubes, reducing heat transfer. That's why your project on Advancements in MSFE Heater Cleaning is important—it focuses on making cleaning faster, safer, and more efficient.

MSFE Project Equipment & Products

Tube Holder Plates

- Function: These plates hold and align the tubes inside the heater shell.
- Use in MSFE: In shell-and-tube heat exchangers, they ensure that heat transfer tubes are securely positioned, allowing efficient steam heating of the feed solution.
- Material: Usually made of corrosion-resistant metals (e.g., SS 316) to withstand hot, corrosive liquids.

Heater Shells and Evaporators

- Function: The heart of MSFE, where feed liquid is heated to cause evaporation.
- Heater Shells: Outer shell where steam is passed to heat the liquid inside tubes.
- Evaporators: Remove water from the feed by boiling it under vacuum, across multiple stages (effects), increasing steam efficiency.
- Material: Built to withstand thermal stress and corrosive environments.

Sand filter

- Function: Used for pre-treatment of feedwater or process liquid.
- Use in MSFE: Removes suspended solids that could cause scaling or fouling inside heater tubes and evaporators.
- Layers: Typically consist of sand, gravel, and pebbles to trap particles.



GRID PLATES



HEATER SHELL



SAND FILTERS



CYCLONE SEPERATOR

- Function: Use centrifugal force to separate vapors from entrained liquid droplets.
- Use in MSFE: After evaporation, these remove any liquid carryover from the vapor stream, ensuring dry steam moves to the next effect or for condensation.
- Efficiency: Prevents loss of product and scaling in downstream equipment.

Star Rollers

- Function: Typically part of a scraper or rotating drum mechanism.
- Use in MSFE: Helps in concentrated slurry handling or crystal discharge from final effects.
- Design: Star-like rotating components to push solids/slurry and prevent blockages.

Intel & Flash Nozzles

- Inlet (Intel) Nozzles: Introduce feed liquid into the evaporator at specific locations to maximize surface contact.
- Flash Nozzles: Inject partially heated feed into a lower pressure effect, causing flash evaporation.
- Goal: Improve phase change efficiency and reduce scaling by directing flow smartly.

Pipes & Fittings

- Function: Carry steam, condensate, feed liquid, and vapor between various components.
- Types: Includes elbows, tees, reducers, valves, etc.
- Material: Must be resistant to corrosion, pressure, and temperature changes (e.g., SS 304, SS 316, FRP-lined pipes).



STAR ROLLERS



FLASH NOZZLE



PIPES AND FITTINGS



GASKETS

- Function: Ensure leak-proof sealing between flanged connections, covers, and joints.
- Use in MSFE: Maintain vacuum integrity and steam-tight seals in heaters and separators.
- Materials: Often made from EPDM, Viton, or PTFE, selected based on chemical compatibility

Research Paper 1 : Cleaning Heater Convection Tubes

Background :
Fired heaters are critical in refineries, but over time, ash, unburned hydrocarbons, and foreign matter build up on the convection tubes, reducing heat transfer. The convection section is difficult to clean due to tight spacing, tube arrangement, and access issues. Liquid fuels cause more deposits than gas fuels, increasing the fouling risk.

Problems:
Narrow gaps between tubes (as little as 30 mm) make mechanical cleaning very hard.
Tube bundles are long (>2 m), and some areas are inaccessible by hand.
Earlier methods involved spraying water and chemicals, which were ineffective for hard-to-reach areas or stubborn deposits.

Advancements in Fired Heater Cleaning Techniques

1. Portable Cleaning Machines : Automated machines with multi-directional rails and high-pressure jets (100 bar) clean large tube surfaces efficiently without scaffolding, reducing manual effort and hazards.
2. Hand Jet Cleaning : Ultra-high-pressure (2700 bar) water jets with flexible nozzles are manually inserted to reach tightly packed or deep tube areas, ensuring thorough cleaning where machines can't reach.
3. Chemical Cleaning Integration : A combination of chemicals is used to dissolve hard deposits and enhance cleaning efficiency :
   Hydrochloric Acid (1–2%) – removes scale and oxides.
   Ammonium Bifluoride – for silica-based or stubborn deposits.
   Surfactants/Detergents – emulsify hydrocarbons and improve cleaning of oily surfaces.
   Corrosion Inhibitors (e.g., PDTDC) – protect metal surfaces during acid exposure.
   Neutralizers (e.g., Sodium Carbonate) – stop acid reaction after cleaning.
   Passivators (e.g., Citric or Nitric Acid) – restore protective oxide layer on metal.
4. Endoscopic Inspection : Industrial cameras are used before and after cleaning to verify effectiveness and guide re-cleaning if needed.
5. Layered Skim Cleaning : Cleaning is done row-by-row in a planned path to ensure uniform coverage across all tube layers.
6. Water Quality Monitoring : Cleaning progress is tracked by observing water clarity (dirty to clear), helping avoid under- or over-cleaning.
7. Optimized Cleaning Sequence : A defined sequence (debris removal => chemicals => jets => rinse => inspection) improves safety, speed, and effectiveness.
8. Performance Monitoring : Post-cleaning analysis

Research Paper 2 : Desalination Plant Acid Cleaning: Corrosion & Inhibition Study

Background :
Elucidating the corrosion characteristics of brine heater and evaporator condenser alloys during acid cleaning of MSF plants and its mitigation. MSF plants suffer from scaling (mineral deposits) on brine heater and evaporator tubes. These scales are removed using acid cleaning, often with 3% HCl at 40 °C for 72 hours. But acid cleaning damages metal components, especially Cu–Ni alloys, unless corrosion inhibitors are used.

Experimental Setup :
Materials studied : CuNi10 (10% nickel), CuNi30 (30% nickel)
New corrosion inhibitor : PDTDC (a nitrogen-containing ionic polymer)
Goal : See if PDTDC can protect these metals under harsh acid cleaning conditions.

Results :

1. Corrosion Behavior
   Both alloys corrode significantly during acid cleaning. CuNi30 showed better resistance than CuNi10 due to higher Ni content. Corrosion led to pitting and surface roughness.
2. Effectiveness of PDTDC
   PDTDC adsorbs onto the metal surface, mainly at cathodic sites. It blocks hydrogen ion (H^+) reduction, slowing down corrosion. It's a mixed-type inhibitor, but cathodic dominant.
3. PDTDC remains effective up to 72 hours and 50 °C, with best results at 48 hours.
4. PDTDC performs well under flow conditions due to improved contact with metal surfaces.

Alloys	Without PDTDC	With 100 mg/L PDTDC	Inhibition Efficiency
CuNi30	1.60 mm/year	0.37 mm/year	77%
CuNi10	18.96 mm/year	8.85 mm/year	53%

Conclusion :
PDTDC is a highly effective and stable corrosion inhibitor for acid cleaning in MSF desalination plants, showing superior performance with CuNi30 and reasonable effectiveness with CuNi10. It serves as a cost-efficient and promising alternative to conventional inhibitors and has potential for use in multi-metal corrosion protection formulations. The study holds strong industrial relevance, especially for desalination units, brine heaters, evaporator cleaning, and systems like MSFE that use Cu–Ni alloys in auxiliary or chemical plants.

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